The Economic Development Premium in Farmland Markets

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Abstract

This paper estimates the development premium that a concentrated voluntary land assembly generates in nearby farmland markets. Using Indiana's voluntary land assembly of more than 8,000 acres for economic development as the empirical setting, we combine parcel-level transaction data from 2018 to 2024 with public records and implement a spatial difference-in-differences design following Currie et al. (2015). Farmland within 5 miles of the assembly site appreciated by approximately 45.5 percent, or $4,246 per acre, relative to a comparison group 10–25 miles away; farmland in the 5–10 mile ring appreciated by 14.9 percent, or $1,390 per acre. For comparison, the urban-fringe development pressure premium across 13 Indiana cities is 10.5 percent. The development shock generates a premium more than four times larger. These findings provide the first systematic evidence that a non-marginal, discrete development demand shock can rapidly and substantially reprice nearby farmland, with implications for farmland affordability, rural land-use planning, and the evaluation of place-based industrial policy.

Keywords: Farmland prices; Development premium; Voluntary land assembly; Regional development shocks; Place-based industrial policy

JEL codes: Q15, R14, R38, H54

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1 Introduction

The existence of a development premium in farmland markets is well documented. Farmland near cities, in fast-growing areas, and along the urban fringe is priced at a premium above its agricultural value, as buyers expect land to eventually transition out of agriculture to higher-value uses (Chicoine, 1981; Plantinga et al., 2002; Livanis et al., 2006; Borchers et al., 2014; Delbecq et al., 2014). These estimates, however, share a common feature: they all capture gradual, spatially diffuse changes in development pressure as urban areas slowly grow outward. Recently, development agencies have assembled mega-sites through individual negotiations with adjacent landowners, often characterized as voluntary land assembly, resulting in a demand shock. The discrete timing and spatial concentration of voluntary land assembly creates a natural experiment which enables credible identification of the development premium, in contrast to the equilibrium estimates of gradual urban encroachment.

Voluntary land assembly has emerged as an increasingly prominent strategy for large-scale regional development across the United States. The shift began with the Supreme Court's decision in *Kelo v. City of New London* (545 U.S. 469, 2005). Although the Court held that economic development could constitute a public use under the Fifth Amendment, the backlash that followed led more than 40 states to restrict eminent domain for economic purposes (Somin, 2009), pushing development agencies toward voluntary parcel-by-parcel assembly to create the mega-site needed for industrial development. Voluntary assembly avoids compulsory acquisition, but creates a different economic problem. Because completing an assembly depends on acquiring every necessary parcel, individual landowners can hold out for a premium, driving acquisition prices well above prevailing market values (Shavell, 2010; Kominers and Weyl, 2012; Brooks and Lutz, 2016). The CHIPS and Science Act of 2022, the Inflation Reduction Act, and a resurgence of prioritizing industrial development in specific geographies, or place-based industrial policy (Kline and Moretti, 2014a; Juhász et al., 2024) have since accelerated the use of voluntary land assemblies. For example, in Michigan, the state spent more than $261 million assembling approximately 1,300 acres near Flint for an advanced manufacturing megasite (Bridge Michigan, 2025, 2026); Tennessee assembled a 3,600 acre site that became Ford's BlueOval City (Tennessee Governor's Office, 2021; Tennessee Lookout, 2021); Ohio committed $67.4 million in state funds to ready a 1,000 acre industrial megasite in Lorain County (WKYC Studios, 2025); and Indiana's Economic Development Corporation assembled more than 6,300 acres of Boone County farmland at nearly $75,000 per acre, totaling more than $475 million in land purchases, and secured contracts to purchase more than 8,000 acres in total, making it one of the largest voluntary land assemblies in recent U.S. history (Arnolt Center for Investigative Journalism, 2025; IEDC, 2026). Further, in the semiconductor sector alone, firms have announced more than 140 projects totaling over $645 billion across 30 states (SIA, 2026), many of which will depend on voluntary land assembly to secure the thousands of contiguous acres these projects require.

Large voluntary assemblies may affect more than the parcels that are directly purchased. When a development agency assembles thousands of acres of farmland at substantially higher prices and publicly commits to industrial development, it does more than transfer land ownership. It creates a concentrated regional development demand shock: a sudden, credible, and spatially localized signal that nearby land may be converted to high-value non-agricultural use. Despite the growing scale of these events across the United States, their consequences for surrounding farmland markets have not been directly examined. In particular, open empirical questions such as whether shocks raise the prices of nearby parcels that are not directly acquired, how far these effects extend, and whether price responses reflect localized development expectations or broader regional spillovers remain unresolved.

This shift in land use demand beyond crop production is important given the role of farmland in the agricultural economy. Farmland accounts for more than 80% of the total value of U.S. farm assets, and plays a central role in farm profitability, credit access, and intergenerational wealth transfer (USDA, 2023). Most prior research has focused on how farmland values are linked to agricultural fundamentals such as crop returns, interest rates, soil productivity, or on non-agricultural determinants like renewable energy development, conservation programs, and urban sprawl (King and Sinden, 1994; Moss, 1997; Livanis et al., 2006; Kuethe et al., 2011; Capozza and Helsley, 2017; Delbecq et al., 2014; Burns et al., 2018; Abashidze and Taylor, 2023; Kunwar, 2024). These studies show that non-agricultural demand is capitalized into land values, but the effects typically reflect gradual and spatially diffuse changes in development pressure. This distinction between gradual development pressure and a concentrated voluntary land assembly shock matters because such a shock can combine a direct demand shift, a credible signal about future industrial development, and changes in local bargaining conditions, all concentrated in space and time.

This study examines how regional development demand shocks affect surrounding farmland prices using the Indiana Economic Development Corporation's (IEDC) acquisition of farmland for the LEAP (Limitless Exploration/Advanced Pace) Innovation and Research District in Boone County, Indiana. Unlike gradual urban development pressure, LEAP represents a discrete, concentrated demand shock. By mid-2024, Indiana Economic Development Corporation (IEDC) had assembled more than 6,000 acres of Boone County farmland at an average price of nearly $72,000 per acre, making it one of the largest state-led voluntary land assemblies in recent U.S. history (Arnolt Center for Investigative Journalism, 2025). The acquisition is geographically concentrated along the I-65 corridor between Indianapolis and Lafayette, creating a clear spatial gradient in exposure to the shock. The public announcement in 2022Q2 provides a clean temporal break for measuring how farmland prices respond to the shock. Section 2 provides institutional background on IEDC and the LEAP project.

Empirically, this study follows Currie et al. (2015), using distance-based comparisons to study how localized shocks are capitalized into property values. We adopt this approach to farmland markets by first documenting the spatial gradient in post-announcement price responses around the LEAP District. This step allows the data to reveal where the price effect is concentrated rather than imposing a treatment radius arbitrarily or assigning treatment at the county level. Based on this gradient, we define three concentric rings around the LEAP site: parcels within 0–5 miles as the localized treatment group, parcels 5–10 miles away as the potential spillover group, and parcels 10–25 miles away as the comparison group. This design is also consistent with recent work on spatial difference-in-differences, which emphasizes that nearby comparison units may themselves be exposed to spillovers and that credible identification requires separating directly treated, potentially exposed, and farther comparison units (e.g. Butts, 2021). We then estimate a difference-in-differences model to quantify the localized and spillover effects, and use an event-study specification to examine dynamic price responses and assess pre-announcement trends. Because the LEAP site was deliberately selected for its locational features, our empirical design does not rely on random siting; instead, identification comes from the timing and spatial concentration of the voluntary land assembly shock within the regional farmland market.

We find that the regional development demand shock generates large, persistent, and spatially structured effects in surrounding farmland markets. Farmland parcels within 5 miles of the LEAP site experienced approximately 45.5 percent price premium, or $4,246 per acre relative to the comparison group following the public announcement. The estimated effect in the 5–10 mile ring is positive but smaller, approximately 14.9 percent or $1,390 per acre, suggesting that price effects may extend beyond the core acquisition zone but decay with distance. Additional event-study estimates provide no clear evidence of differential pre-announcement trends, followed by a sharp rise in localized price premiums after the public announcement.

We also estimate a pooled cross-sectional hedonic model across 13 Indiana cities and find an urban-fringe proximity premium of 10.5 percent, consistent with prior equilibrium benchmarks (Plantinga et al., 2002; Livanis et al., 2006). Plantinga et al. (2002) estimate that approximately 8 percent of Indiana farmland values reflect expectations of future land development, while Livanis et al. (2006) and Borchers et al. (2014) find positive but modest relationships between farmland values and urban accessibility or development potential. The estimate from the LEAP district of 45.5 percent is more than four times larger, providing direct evidence that a concentrated regional development demand shock can rapidly reprice nearby farmland well beyond what equilibrium development pressure alone would predict.

The contribution of this paper is not simply to document another form of development pressure, but to show that a discrete, credible, and spatially concentrated regional development shock can rapidly reprice nearby farmland. This premium reflects how an unanticipated development shock shifts demand, updates expectations about future land use, and alters bargaining conditions in a localized area. These findings have implications for land-use planning, farmland affordability, and the design of voluntary assembly programs, particularly as regional development initiatives become an increasingly prominent feature of place-based industrial policy across the United States (Kline and Moretti, 2014b; Neumark and Simpson, 2015; U.S. Government Accountability Office, 2021). Project-level cost-benefit analyses that focus on acquired parcels, jobs, and infrastructure investment may therefore understate the broader land-market consequences of regional development policy. As states compete for industrial development through voluntary land assembly, policymakers must weigh these benefits against effects on rural land markets and consider how the costs of development are distributed across landowners, farmers, and other rural stakeholders.

The remainder of the paper is organized as follows. Section 2 provides institutional background of the Indiana Economic Development Corporation (IEDC) and the LEAP Research and Innovation District. Section 3 outlines the conceptual framework and empirical strategy. Section 4 presents the data and reports descriptive statistics. Section 5 reports the results and discusses their economic magnitude and interpretation. Finally, Section 6 provides a concluding summary of key findings.

2 Background Information

The Indiana Economic Development Corporation (IEDC) is Indiana's lead economic development quasi-public agency. It was established in 2005 to replace the Indiana Department of Commerce and is designed to streamline Indiana's economic development strategy by consolidating programs and giving the state more flexibility in competing for business investment. The primary objectives of IEDC are to attract and retain firms, facilitate workforce development, coordinate infrastructure, and prepare "shovel-ready" sites for large-scale projects. Unlike private buyers in farmland markets, IEDC has both the financial resources and the political authority to voluntarily assemble farmland at scale for non-agricultural purposes, often purchasing parcels at prices above prevailing market values to secure land for economic and industrial development. In late 2021, IEDC began acquiring farmland in Boone County, Indiana, and in May 2022 formally announced its most ambitious land-based initiative to date: the LEAP (Limitless Exploration/Advanced Pace)

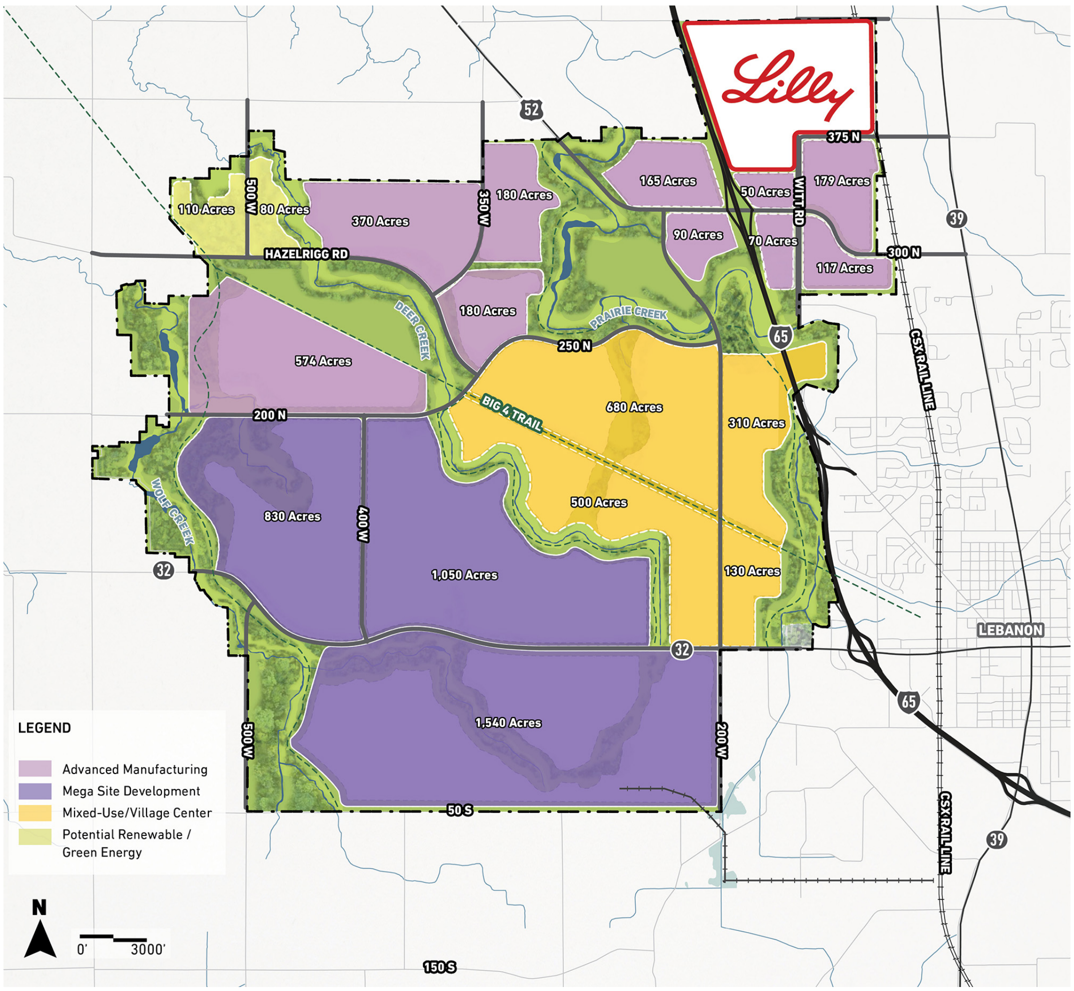
Innovation and Research District.

The LEAP Innovation and Research District is a large-scale industrial and research hub situated along the I-65 corridor between Indianapolis and Lafayette. The project is designed by IEDC to attract firms in advanced manufacturing, life sciences, semiconductors, clean energy, and other technology-intensive sectors (IEDC, 2026). State officials have promoted LEAP as a transformative ”mega-site” capable of positioning Indiana as a national leader in high-tech investment (IEDC, 2026). By March 2023, IEDC had spent approximately $126 million to acquire more than 1,500 acres, with the long-term goal of creating an 11,000-acre high-tech innovation park. The purchase price averaged roughly $72,000 per acre, approximately ten times the Indiana average cropland value of $7,170 per acre (USDA NASS, 2022). By mid-2024, IEDC reported ownership or contracts for more than 8,000 acres, making it one of the largest state-led voluntary land assemblies in recent U.S. history (Indiana Business Journal, 2024).

Eli Lilly and Company was the project’s first anchor tenant; as part of a larger commitment to invest $18 billion in Indiana facilities, Eli Lilly announced plans to build a pharmaceutical manufacturing campus worth $3.7–4.0 billion after purchasing 605 acres from IEDC for $60 million in April 2023 (Indiana Capital Chronicle, 2025). The State Budget Committee subsequently approved IEDC an additional $88 million in June 2024 to expand the district, allowing acquisition of another 1,400 acres with 1,100 acres under contract. IEDC expects the project to attract additional high-tech tenants and generate roughly 10,000 new jobs (Indiana Business Journal, 2024). The geographic scope of farmland acquisition is shown in Figure 1, which maps parcels purchased or contracted by IEDC for the LEAP District.

The LEAP District has also generated substantial local scrutiny, particularly around its environmental and land market implications. Residents and county officials have raised concerns regarding the project’s significant water need and potential ecological risks, including proposed pipelines drawing water from the Wabash River basin (Indianapolis Business Journal, 2023). In addition, IEDC has been purchasing farmland at prices far above prevailing agricultural value, sometimes eight to ten times higher than the state average. This higher-price land purchase has raised concerns that such transactions may reset price benchmarks and distort land price expec-

Figure 1: Farmland acquired or under contract by IEDC for the LEAP District, Boone County, Indiana (2024).



Notes: The figure above illustrates farmland parcels acquired or placed under contract by the Indiana Economic Development Corporation (IEDC) to assemble the LEAP District site. As of mid-2024, IEDC reported control of approximately 8,000 acres through direct purchases and pending contracts. This map is obtained from the IEDC website.

tations beyond the parcels directly acquired (Arnolt Center for Investigative Journalism, 2025; Indiana Capital Chronicle, 2023).

For this study, the LEAP District provides an empirical setting for examining how a concentrated regional development demand shock is capitalized into surrounding farmland prices. Before IEDC's intervention, farmland prices in the study area were shaped by similar regional agricultural fundamentals, including soil productivity, commodity-market conditions and broader Indiana farmland demand. The sudden and geographically concentrated entry of a well-capitalized institutional buyer therefore provides a clear external shock to this market. Because the acquisition is spatially concentrated, exposure to the shock varies continuously with distance from the LEAP site rather than simply across county boundaries. This spatial gradient in exposure motivates the distance-based identification strategy used in this study, while the parcel-level event-study analysis formally assesses pre-announcement trends.

3 Conceptual Framework / Identification Strategy

3.1 Conceptual Framework

Farmland prices reflect both the value of land in agricultural production and the option value associated with future non-agricultural use (Plantinga et al., 2002; Wrenn and Irwin, 2015). This distinction is especially important for understanding how a voluntary land assembly shock propagates through surrounding markets. Voluntary land assembly does not simply remove parcels from agricultural production but it also changes expectations about future industrial development in the neighboring land market. To capture these channels, we develop a partial equilibrium model of farmland pricing, following the approach of Currie et al. (2015), in which land values incorporate both agricultural returns and development option value. As in Currie et al. (2015), this model motivates our empirical strategy and generates testable predictions about the spatial distribution and timing of price responses following a concentrated regional development demand shock.

Consider a local land market consisting of parcels distributed across two locations, $g \in N, F$, defined by proximity to the development site: parcels near the site ($g = N$) and parcels farther away ($g = F$). Following the land conversion literature (Plantinga et al., 2002; Wrenn and Irwin, 2015), the price of parcel i in location g at time t reflects two sources of value:

$$P_{igt} = A_{it} + \pi_g \cdot \Omega_t + \varepsilon_{igt}. \tag{1}$$

The term A_{it} captures the agricultural use value of the parcel, including soil quality, parcel size, cropland share, improvements, and other characteristics that determine expected returns from farming. The second term $\pi_g \cdot \Omega_t$ captures development option value. Here, $\pi_g \in [0, 1]$ is the location-specific probability that the parcel in location g will be converted to, acquired for, or otherwise affected by future development, and Ω_t is the expected net surplus from development relative to continued agricultural use. Prior to voluntary land assembly shock, parcels near the development site may already have contained some option value, as most regional land markets are not isolated from broader growth pressures. However, a concentrated regional development demand shock plausibly increases both the perceived probability and scale of future development near the site. This shock fundamentally alters this equilibrium by credibly signaling a large positive development value $D_t > 0$ and demonstrating that parcels near the site face a strictly positive conversion probability $\pi_N > 0$.

Economic theory and prior land-market research suggest that such a regional development demand shock can affect farmland prices through three distinct channels. The first channel is a direct demand effect. In a market where transaction volumes are low and prices are sensitive to changes in demand, the entry of a large institutional buyer can shift local price expectations and create new benchmarks for subsequent arm's-length transactions. This channel is especially relevant when the buyer is motivated by long-term development objectives rather than agricultural returns. The second channel operates through development pressure and expectations. The literature on speculation and development expectations shows that farmland values capitalize not only current agricultural rents but also the expected probability of future land conversion (Shi et al., 1997; Plantinga et al., 2002; Capozza and Helsley, 2017). A credible public announcement and infrastructure commitments associated with the development project can therefore increase farmland prices even for parcels that are never directly purchased, as market participants revise upward their beliefs about future land use. Because projected returns to industrial or commercial development often far exceed returns from row-crop production (Capozza and Helsley, 2017; Feng and Hayes, 2020) and because such transitions are frequently viewed as fiscally beneficial through an expanded property tax base (Neumark and Simpson, 2015), even a modest increase in perceived conversion probability can produce large price responses. Third, the presence of a well-capitalized institutional buyer alters market structure and bargaining positions. In thin land markets, transaction prices are often determined by bilateral negotiation rather than competitive price-taking (King and Sinden, 1994; Harding et al., 2003). When a well-capitalized buyer assembles land at scale, sellers revise their reservation prices upward due to the holdout problem, while competing buyers' bargaining power becomes weaker (Kominers and Weyl, 2012; Shavell, 2010; Brooks and Lutz, 2016). Moreover, sellers who receive above-market acquisition may re-enter the regional market as buyers under Section 1031 like-kind exchange provisions, bidding up prices beyond the immediate acquisition zone and propagating elevated reservation prices outward[1] (Dillard et al., 2013). Together, these channels operate through π_g and Ω_t in equation 1 and share the common prediction that price effects are largest where development probability is highest, parcels immediately adjacent to the source of the regional development demand shock.

The empirical comparison follows the logic that parcels close to the development shock and parcels farther away are exposed to many of the same regional farmland-market conditions, including commodity prices, interest rates, statewide farmland demand, and broader macroeconomic shocks. However, parcels close to the shock should receive the strongest localized development-value shock. This dynamic is sometimes called *near-far* logic. Let N denote parcels near the LEAP site and F denote farther parcels within the same broader regional farmland market. To formalize identification, let τ represent the public announcement date of the voluntary land assembly. Taking pre-to-post first differences of equation 1 within each location:

[1] Section 1031 of the U.S. Internal Revenue Code allows property owners to defer capital gains taxes on the sale of real property by reinvesting the proceeds into a like-kind replacement property. The seller must identify a replacement property within 45 days and complete the exchange within 180 days. This time pressure creates strong incentives to reinvest quickly, often at above-reservation prices, which can elevate bids in the surrounding land market.

$$\Delta P_{ig} = \Delta A_i + \Delta(\pi_g \cdot \Omega) + \Delta\varepsilon_{ig}, \tag{2}$$

where Δ denotes the change between pre- and post-announcement periods. The first term captures the change in agricultural value driven by shifts in commodity returns and interest rates, factors that are common to all farmland regardless of proximity to the development site. The second term captures the emergence of location-specific development option value following the shock. Under the parallel trends assumption, that in the absence of the IEDC LEAP District, farmland prices in near and far locations would have followed the same trajectory, differencing equation 2 across locations eliminates the common agricultural component and yields:

$$\hat{\beta} = \Delta\bar{P}_N - \Delta\bar{P}_F = (\pi_N - \pi_F) \cdot \Delta\Omega \tag{3}$$

Equation 3 shows that the spatial difference-in-differences estimator recovers the differential increase in development option value for parcels closer to the development site relative to parcels farther away. In settings with spatial spillovers, nearby comparison units may be partially exposed to treatment, so separating directly exposed parcels, potentially exposed spillover parcels, and farther untreated comparison parcels is important for credible identification (Butts, 2021). Because proximity should raise the perceived probability of development more for nearby parcels than for farther parcels, $\pi_N > \pi_F$. Therefore, if the development shock increases expected development surplus, $\Delta\Omega > 0$, the predicted treatment effect is positive.

A central implication of the framework is that the treatment effects should be strongest close to the source of the development demand shock and weaker farther away. The perceived probability of development, π_g, is a decreasing function of distance from the development site. Letting d denote distance from the shock site, and differentiating equation 3 with respect to distance yields $\frac{\partial \pi(d)}{\partial d} < 0$. Parcels closer to the site are more likely to be affected by infrastructure expansion, and complementary industrial development. So, the development premium capitalized into farmland prices should therefore be greater near the shock site and decline as parcels become less connected to the planned development area. This spatial prediction guides the empirical design. Rather than imposing arbitrary distance cutoffs, we first estimate how price changes vary across narrow distance bins around the development site. The spatial gradient shows where the price response is concentrated and where it becomes small enough for parcels to serve as a plausible comparison group. In the second step, we use these first-stage estimates to define the localized treatment region, potential spillover region, and comparison region for the main difference-in-differences analysis. This approach follows (Currie et al., 2015), who use an analogous spatial gradient strategy to identify the effects of industrial plant openings on housing values and infant health outcomes.

In the Indiana LEAP setting, these arguments generate two testable predictions. First, farmland prices should increase most sharply for parcels closest to the LEAP site and decline with distance. Second, price effects in areas beyond the immediate acquisition zone may be positive but smaller in magnitude, reflecting broader revaluation of nearby land-market expectations. The empirical analysis centers on the public LEAP announcement and does not attempt to separately identify pre-announcement information flows or private bargaining dynamics.

3.2 Identification Strategy

To estimate the localized price effects of the LEAP regional development demand shock on farmland prices, we implement a three-step spatial difference-in-differences strategy following Currie et al. (2015). First, we estimate the spatial gradient of price responses around the LEAP site to determine empirically where the effects concentrate. Second, we use the data-driven gradient obtained from the first step to define treatment, spillover, and comparison groups for the main difference-in-differences model. Finally, we estimate an event study that examines timing and pre-treatment dynamics.

3.2.1 Spatial Gradient

One concern with ring-based DiD designs is that the relevant treatment boundary is not predetermined. Researchers often impose treatment rings using arbitrary distance cutoffs, but misspecified boundaries can either bias the estimated treatment effect or conflate direct effects with spillovers. We address this concern by first estimating the spatial gradient of the price response before imposing any ring structure. Specifically, we partition all transactions into distance bins of varying width: 0–1, 1–3, 3–5, 5–10, 10–15, 15–20, and 20–25 miles from the nearest parcel acquired by IEDC. We then estimate:

$$\log P_{igt} = \alpha + \sum_{b \neq 20\text{–}25} \lambda_b (\mathbf{1}[\text{bin}_i = b] \times Post_t) + \mathbf{X}'_{igt}\boldsymbol{\gamma} + \mu_c + \tau_t + \varepsilon_{igt} \tag{4}$$

In the equation 4, 20–25 mile bin is the reference category. The coefficients $\{\lambda_b\}$ trace the price effect of LEAP District shock as a continuous function of distance without imposing any boundary. We use the distance at which λ_b becomes economically negligible and statistically indifferent from zero to determine the treatment and spillover ring boundaries used in subsequent steps. The estimated gradient is presented in Section 5.

3.2.2 Difference-in-Differences

The second step uses the spatial-gradient results to classify parcels into three concentric rings. Based on the estimated distance pattern, parcels within 0–5 miles of the LEAP district are classified as the *treatment ring*. These groups contain parcels subject to direct demand pressure, speculative revaluation, and bargaining spillovers. Similarly, parcels that are exposed to broader market revaluation but unlikely to be directly acquired are classified as *spillover ring*, and are 5-10 miles away from LEAP district. Parcels 10–25 miles away serve as the counterfactual or the comparison group, sharing the same regional market conditions but lying beyond the spatial reach of the designation shock. The estimating equation is:

$$\log P_{igt} = \alpha + \beta_1 Treat_i + \beta_2 Spill_i + \delta_1 (Treat_i \times Post_t) + \delta_2 (Spill_i \times Post_t) + \mathbf{X}'_{igt}\boldsymbol{\gamma} + \mu_c + \tau_t + \varepsilon_{igt} \tag{5}$$

where $Treat_i$ equals one for parcels within 0–5 miles of the LEAP district, and $Spill_i$ equals one for parcels 5–10 miles away. The comparison group is parcels 10–25 miles from the site. $Post_t = \mathbf{1}[t \geq 2022Q2]$ corresponds to the public announcement of the LEAP project in the second quarter of 2022. The coefficients δ_1 and δ_2 are the parameters of interest, measuring the price effect on treatment and spillover parcels relative to the comparison ring. The vector $\mathbf{X}_{igt}$ includes additional controls used in the estimation. It includes transaction acreage, soil productivity (NCCPI score), improvement, share of cropland, and an indicator for corporate buyers. To isolate the treatment effect from unrelated spatial amenities, we additionally control for distances to substations, utility-scale solar installations, transmission lines, wind turbines, rail lines, ethanol plants, and Indianapolis.

Equation 5 mirrors the hedonic difference-in-differences estimator in Currie et al. (2015), who regress log housing values on distance-ring indicators interacted with post-exposure dummies to recover the capitalization of environmental disamenities into property prices. In our study, regional development demand shock plays the role of the amenity shock, and agricultural land transactions replace housing sales. Following the decomposition in Equation 3, these recover the change in expected development premiums $\Delta\Omega$ scaled by the ring-specific development probability, π_N and π_F respectively. County fixed effects μ_c absorb time-invariant local market conditions, and year fixed effects τ_t absorb common price trends across all rings. Standard errors are clustered at the county level.

Identification requires that, absent the LEAP shock, prices in the localized treatment and spillover regions would have followed the same trend as prices in the comparison region, conditional on observed controls, county fixed effects, and year fixed effects. This design follows the same near–far strategy as Currie et al. (2015), where locations closer to a localized shock are compared with locations farther away but still in the same broader regional land market. The key identifying assumption is that farmland within the 0–5 mile and 5–10 mile rings would not have experienced a discrete post-announcement price increase relative to farmland 10–25 miles away in the absence of the LEAP project. We assess this assumption using the event-study specification below.

A limitation of this setting is that the LEAP site was not randomly selected. IEDC chose Boone County because of its access to I-65, proximity to Indianapolis and Purdue University, existing infrastructure, and availability of large agricultural parcels. These locational features may also be correlated with farmland price growth. Our Difference-in-Differences model therefore does not rely on random siting of LEAP District. Instead, identification comes from the timing and spatial concentration of the development demand shock. The key comparison is between parcels located closer to the LEAP District and parcels farther away within the same regional land market, conditional on parcel characteristics, agricultural quality, infrastructure access, county fixed effects, and year fixed effects. This strategy asks whether farmland closest to the LEAP site experienced a discrete post announcement premium relative to nearby comparison parcels, rather than whether Boone County as a whole was randomly exposed to development pressure.

3.2.3 Event Study

To test the parallel trends assumption and characterize the timing of price effects, we estimate an event-study extension of Equation (5). The event-study equation is:

$$\log P_{igt} = \alpha + \sum_{k \neq -1} \beta_k^N (\mathbf{1}[t - t^* = k] \cdot Treat_g) + \sum_{k \neq -1} \beta_k^F (\mathbf{1}[t - t^* = k] \cdot Spill_g) + \mathbf{X}'_{igt}\boldsymbol{\gamma} + \mu_c + \tau_t + \varepsilon_{igt} \tag{6}$$

where $t^* = 2022Q2$ is the LEAP announcement quarter and $k = -1$, corresponding to 2022Q1, is the reference period. The coefficients $\{\beta_k^N\}$ trace the quarterly price path for parcels in the localized treatment region relative to the comparison region, while $\{\beta_k^F\}$ trace the corresponding path for parcels in the spillover region. The coefficients are interpreted relative to the omitted quarter, so the pre-treatment estimates show whether treated and spillover parcels were already experiencing differential price growth before the LEAP project became public.

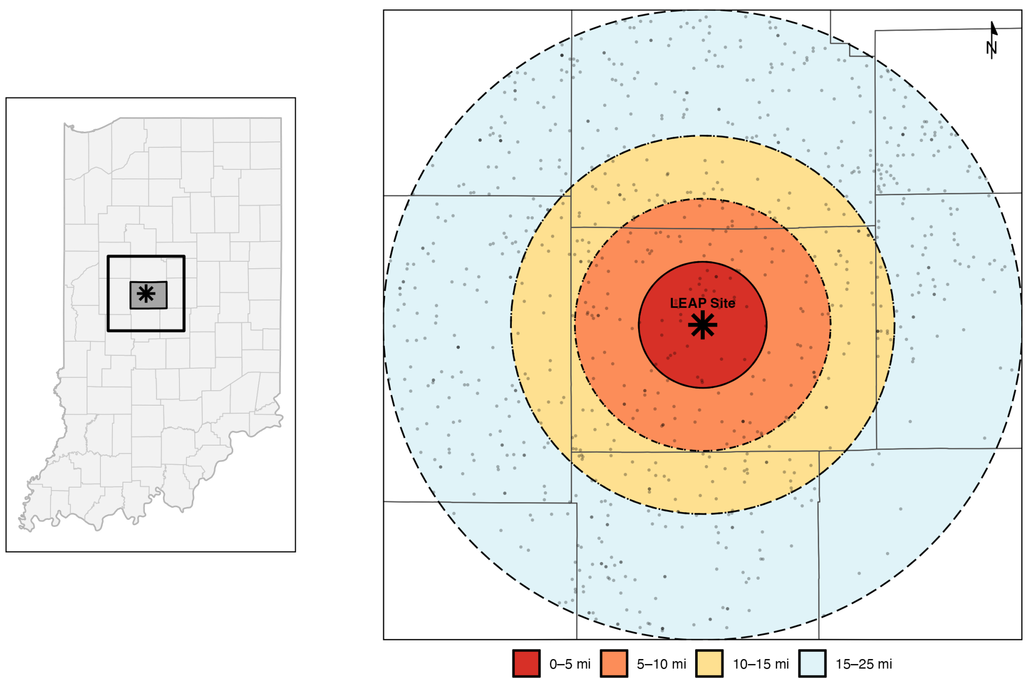
4 Data

The primary objective of this study is to examine how a concentrated regional development demand shock is capitalized into surrounding farmland prices. To do this, we analyze the IEDC farmland acquisitions for the LEAP District in Boone County, Indiana, and their price effects on nearby non-acquired parcels. The farmland transactions data used in this study are obtained from proprietary database of Acres.com, a digital farmland intelligence platform that compiles property deed transfers, assessor records and geospatial parcel data into a standardized format. This database is restricted to agricultural and farmland transactions, ensuring that commercial, industrial, and residential transfers are excluded by construction. Our sample covers all farmland transactions within 25 miles of the LEAP site from 2018 through 2024. This transaction data includes the total sale price, date of transaction, acreage, National Commodity Crop Productivity Index (NCCPI), improvement (an indicator for the presence of structures), the share of cropland in the transaction at the time of sale, an indicator for corporate buyers and geospatial attributes like distance to nearest electricity substations, solar farm, wind turbines, transmission lines, rail lines, and ethanol plants. The database further provides parcel-level geospatial coordinates, which we use to compute distances to the IEDC LEAP district.

To geolocate the LEAP site and identify parcels acquired by IEDC, we obtain a dataset of IEDC land acquisitions from the Boone County Assessor's Office Sales Disclosure Form (SDF) records, which document all deed transfers recorded in the county. These records provide transaction prices, sales dates, parcel identifiers, and geospatial coordinates for each land acquisition, which are used in this study to compute a parcel distance variables. The IEDC acquisition data indicate that the agency assembled approximately 6,156 acres for the LEAP district, with total purchase value of about $446 million. This data feature implies an acreage-weighted average purchase price of roughly $72,000 per acre, approximately ten times the Indiana average cropland value of $7,170 per acre in 2022 (USDA NASS, 2022). Our final analysis sample exclude all of these parcels acquired by IEDC as these transactions reflect institutional acquisition prices well above prevailing market levels and would conflate the treatment shock with the outcome variable. Further, we restrict the sample to arm's-length transactions by excluding sales where the buyer and seller share the same name. The resulting analysis sample covers arm's-length farmland transactions from 2018 through 2024.

Treatment assignment is geographically determined following the two-step spatial identification process as explained in Section 3.2. Rather than assigning treatment at the county level, we exploit the spatial gradient of price effects following Currie et al. (2015) to classify each parcel by its distance from the LEAP site into concentric rings. The treatment period begins in 2022Q2, when the LEAP District was publicly announced. Transactions from the fourth quarter of 2021, when IEDC started land purchases prior to public disclosure, are also included in the sample and examined in the event study as part of the pre-announcement period. The study area and ring boundaries are presented in Figure 2.

Figure 2: Study Area



Notes: The left panel locates Boone County within Indiana. The right panel shows the concentric distance rings centered on the IEDC/LEAP acquisition district: treatment (0–5 miles), spillover (5–10 miles), and comparison (10–25 miles). Points represent arm's-length farmland transactions in the analysis sample.

Table 1 reports summary statistics for the major variables in the analysis, disaggregated by distance ring from the LEAP site. Average farmland prices are highest in the 5-10 mile ring ($10,803 per acre) and decline with distance to $9,331 per acre in the 10-25 mile ring.[2] Soil productivity (NCCPI) is nearly identical across all rings, averaging between 76.2 and 77.3, indicating that the rings are comparable in underlying agricultural quality. This similarity supports the use of outer-ring transactions as the counterfactual for inner-ring transactions in the difference-in-differences design. Distance to wind turbines increases sharply with proximity to the site, reflecting the concentration of wind energy development in northern Indiana away from the LEAP location [3]

[2]We find that the unconditional mean price per acre is slightly higher in 5–10 mile ring relative to the 0–5 mile. These raw means exclude IEDC acquisition transactions and non-farm transactions, so the comparison reflects open-market, arm's-length farmland sales. Differences in average prices likely reflect compositional differences in parcel characteristics. The regression analysis accounts for these differences using parcel-level controls, county fixed effects, and year fixed effects. More importantly, the difference-in-differences estimator is identified from changes in prices within each ring over time relative to the comparison ring, rather than from cross-sectional differences in price levels across rings. Thus, higher average prices in one ring do not by themselves imply a treatment effect.

[3]We control directly for wind turbine and all other distance-based amenities in the regression specifications.

Table 1: Summary Statistics by Distance from the IEDC/LEAP Footprint

Mean (SD)				
Variable	Full sample	0–5 mi	5–10 mi	10–25 mi
Price per acre ($)	9,618.17	10,172.33	10,802.76	9331.1
	(5,562.97)	(5,675.85)	(6,288.20)	(5,376.96)
Parcel characteristics				
Acres	67.93	74.18	64.45	67.77
	(57.07)	(96.54)	(49.50)	(51.38)
NCCPI	76.43	76.86	77.24	76.23
	(9.85)	(6.70)	(7.53)	(10.54)
Has improvement	0.15	0.12	0.14	0.16
	(0.36)	(0.33)	(0.35)	(0.37)
Cropland share	0.79	0.84	0.84	0.77
	(0.29)	(0.24)	(0.23)	(0.31)
Buyer characteristics				
Corporate buyer	0.35	0.34	0.37	0.35
	(0.48)	(0.48)	(0.48)	(0.48)
Distance controls				
Distance to Indianapolis (mi)	34.05	26.67	28.38	36.02
	(11.66)	(4.75)	(8.13)	(12.11)
Distance to substation (mi)	0.80	0.68	0.87	0.81
	(0.45)	(0.36)	(0.51)	(0.45)
Distance to solar farm (mi)	2.87	3.63	3.35	2.68
	(1.81)	(2.02)	(1.78)	(1.75)
Distance to transmission line (mi)	0.77	0.66	0.71	0.80
	(0.47)	(0.38)	(0.37)	(0.49)
Distance to wind turbine (mi)	8.25	14.87	11.06	6.89
	(5.81)	(3.66)	(5.26)	(5.34)
Distance to railroad (mi)	0.84	1.00	0.96	0.79
	(0.56)	(0.60)	(0.59)	(0.54)
Distance to ethanol plant (mi)	8.73	6.35	9.12	8.96
	(4.38)	(3.06)	(4.28)	(4.45)
Observations	1,196	116	167	913
Total acres transacted	81,241	8,605	10,762	61,873

Notes: The table reports summary statistics of farmland transactions within 25 miles of the IEDC/LEAP acquisition site. Standard deviations are reported in parentheses. Rings are defined by each transaction's distance to the nearest parcel in the IEDC/LEAP acquisition site. Has improvement, cropland share, and corporate buyer are reported as proportions. Price per acre is reported in nominal dollars.

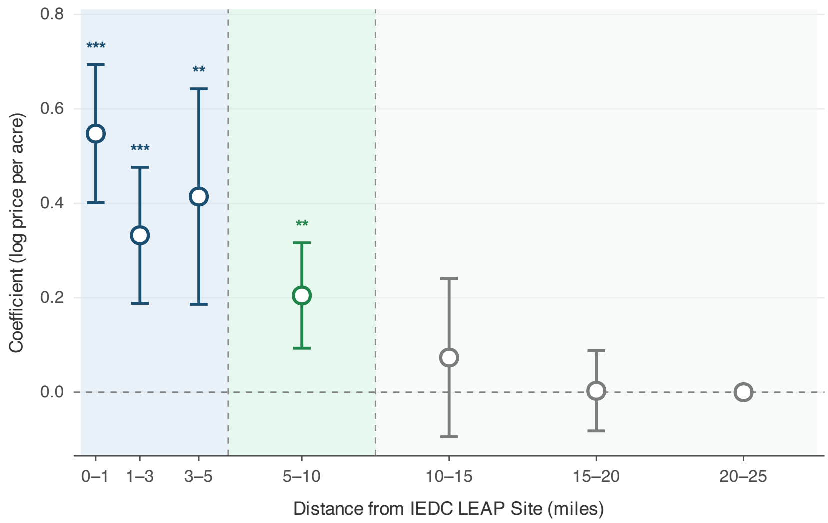
5 Results and Discussion

This section provides empirical evidence on how the LEAP regional development demand shock affects surrounding farmland prices. Results are presented in three steps following the identification strategy in Section 3.2. We first examine the spatial gradient of price effects across distance bins to motivate and validate the ring boundaries used in the analysis. The relevant treatment boundary is not predetermined but instead inferred from the data. We then use these ring boundaries to present the difference-in-differences estimates of the localized and spillover effects and discuss the mechanisms driving them. Finally, we trace the dynamic price response through event-study estimates to test the parallel trend assumption and provide evidence on the timing of market adjustment.

Figure 3 reports the estimated spatial gradient in post-2022 farmland price changes, derived from the Equation 4. The gradient reveals a clear pattern of distance decay: the price premium is largest and statistically significant for parcels within 5 miles of the site, declines sharply in the 5–10 mile bin, and becomes indistinguishable from zero beyond 10 miles. This monotonically declining pattern is consistent with Prediction 1 from Section 3.1, which predicts that the price effect decline with geographic distance as the probability of direct development impact and the intensity of bargaining spillovers diminish. Based on this empirical gradient, we define three concentric rings for the difference-in-differences analysis: a treatment ring (0–5 miles) capturing the zone of largest price premiums, a spillover ring (5–10 miles) where effects are positive but sharply declining as compared to treatment ring, and a comparison ring (10–25 miles) where no statistically significant price response is detected. This data-driven ring definition ensures that the boundaries are not chosen to maximize treatment effects, but instead reflect the actual spatial reach of the development demand shock.

Table 2 reports the difference-in-differences estimates across multiple specifications, with parcel-level controls, fixed effects, and distance controls for spatial attributes. Across all specifications, the localized treatment effect, the coefficient on the interaction between the 0–5 mile ring and the post-2022 period, is remarkably consistent, ranging from 0.37 to 0.4 log points. In the pre-

Figure 3: Spatial Gradient Result



Notes: Each point plots the estimated post-announcement price premium for a given distance bin from Equation 4, relative to the 20–25 mile reference bin. Vertical bars show 95% confidence intervals based on county-clustered standard errors. The figure motivates the ring boundaries used in the difference-in-differences analysis: price effects are large and significant within 5 miles, positive but declining in the 5–10 mile bin, and indistinguishable from zero beyond 10 miles.

ferred specification (Model 5), which includes the full set of controls and county and year fixed effects, the treatment ring coefficient is 0.375 log points, corresponding to an approximately 45.5 percent increase in farmland prices within 5 miles of the IEDC site relative to the comparison ring. [4] This treatment effect translates to approximately $4,246 per acre.[5]

This large and persistent localized price increase reflects all the mechanisms outlined in Section 3.1, but it should not be interpreted as directly identifying each mechanism separately. First, IEDC's land buying at prices well above prevailing market values shifts the local demand curve outward and reset price benchmarks for subsequent arm's-length transactions. Second, the pub-

[4] In a log-linear model, a coefficient $\hat{\beta}$ implies a percentage change of $(e^{\hat{\beta}} - 1) \times 100$. For $\hat{\beta} = 0.375$: $e^{0.375} - 1 = 0.455$, or 45.5 percent.

[5] The dollar magnitude is computed as $\bar{P}_{\text{comparison}} \times (e^{\hat{\beta}} - 1)$, where $\bar{P}_{\text{comparison}} = \$9,331$ per acre is the mean price in the comparison ring (10–25 miles) over the sample period.

Table 2: Effect of IEDC Regional Development Demand Shock on Farmland Prices

	Dependent variable: log(Price per Acre)				
	(1) OLS	(2) +Year FE	(3) +County FE	(4) +Controls	(5) +Infrastructure
A. Treatment effects					
Treatment × Post	0.402***	0.403***	0.401***	0.398***	0.375***
	(0.073)	(0.083)	(0.078)	(0.070)	(0.073)
Spillover × Post	0.116	0.112	0.128*	0.151**	0.139**
	(0.077)	(0.074)	(0.064)	(0.063)	(0.059)
B. Pre-treatment levels					
Treatment	−0.019	−0.019	0.031	0.025	0.087
	(0.045)	(0.052)	(0.035)	(0.054)	(0.083)
Spillover	0.096	0.098	0.075	0.052	0.076
	(0.069)	(0.068)	(0.083)	(0.090)	(0.092)
Post 2022	0.248***				
	(0.048)				
C. Controls					
Log(Acres)				−0.187***	−0.185***
				(0.033)	(0.032)
Soil Quality (NCCPI)				0.001	0.000
				(0.002)	(0.003)
Has Improvement				−0.123**	−0.124**
				(0.043)	(0.042)
Percent Cropland				0.093	0.107
				(0.111)	(0.112)
Corporate Buyer				0.197***	0.189***
				(0.056)	(0.052)
Dist. to Indianapolis (mi)				0.001	0.002
				(0.005)	(0.004)
Infrastructure controls					
Dist. to Substation (mi)					0.020
					(0.057)
Dist. to Solar Farm (mi)					−0.040***
					(0.011)
Dist. to Transmission (mi)					−0.034
					(0.040)
Dist. to Wind Turbine (mi)					0.006
					(0.009)
Dist. to Railroad (mi)					−0.018
					(0.035)
Dist. to Ethanol Plant (mi)					0.009
					(0.007)
Year FE	No	Yes	Yes	Yes	Yes
County FE	No	No	Yes	Yes	Yes
Observations	1,196	1,196	1,196	1,196	1,196
R^2	0.058	0.068	0.103	0.146	0.151

Notes: Comparison group comprises farmland transactions 10–25 miles from IEDC purchase sites. In the preferred specification (Model 5), farmland within 5 miles of the LEAP site appreciated by approximately 45.5 percent ($4,246 per acre) relative to the comparison group, while parcels in the 5–10 mile spillover ring appreciated by approximately 14.9 percent ($1,390 per acre). Standard errors clustered by county in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

lic announcement of large scale economic development projects may have triggered speculative land price revaluation, as market participants updated expectations about the probability of future land conversion. Third, the entry of a well-capitalized buyer and a clear development objective strengthens sellers' bargaining positions and raises reservation prices throughout the treatment zone. While the direction of the estimated effects is consistent with these three mechanisms, the transaction data do not allow us to directly observe or decompose these channels. We therefore interpret the mechanism evidence as suggestive rather than definitive. Table A1 reports heterogeneity tests based on proximity to actual IEDC-acquired parcels, similarity to acquired parcels,[6] and corporate-buyer transactions. The localized price effect is larger for parcels near actual IEDC purchases, parcels similar to IEDC acquisition targets, and transactions involving corporate buyers. These patterns are consistent with the directions predicted by acquisition-proximity, development-expectations, and market-structure channels, respectively. However, because the proxy variables are not direct or exclusive measures of each mechanism, we interpret the estimates as suggestive rather than a full decomposition of mechanisms.[7]

The estimated spillover effect in the 5–10 mile ring is positive but smaller and less robust to alternative inference procedures than the localized treatment effect. In the preferred specification (Model 5), the coefficient on Spillover × Post is 0.139 log points, corresponding to a 14.9 percent increase in farmland prices relative to the comparison ring.[8] The implied spillover premium is approximately \$1,390 per acre.[9] Consistent with Prediction 2 from Section 3.1, price effects are positive but smaller beyond the immediate acquisition zone. This spillover premium is roughly one-third the magnitude of the treatment effect, suggesting that development-related expectations may extend outward from the treatment zone, although this evidence is weaker than the localized treatment effect. Landowners in 5–10 mile ring observe the elevated realized prices in transactions closer to the site, and may revise their reservation prices upward. That is, they benefit from expec-

[6]We define a parcel as *similar to IEDC acquisition targets* if it lies above the sample median in both tract acreage and NCCPI soil-quality score, reflecting the observed characteristics of parcels the IEDC actually purchased.

[7]The benchmark estimate draws on 28 post-announcement parcels located within one mile of an IEDC-acquired parcel; the corporate-buyer estimate draws on 17 post-announcement corporate transactions in the treatment ring. Neither proxy variable exclusively captures its proposed channel.

[8]Following the same transformation as for the 0–5 mile ring, $e^{0.139} - 1 = 0.149$, or 14.9 percent.

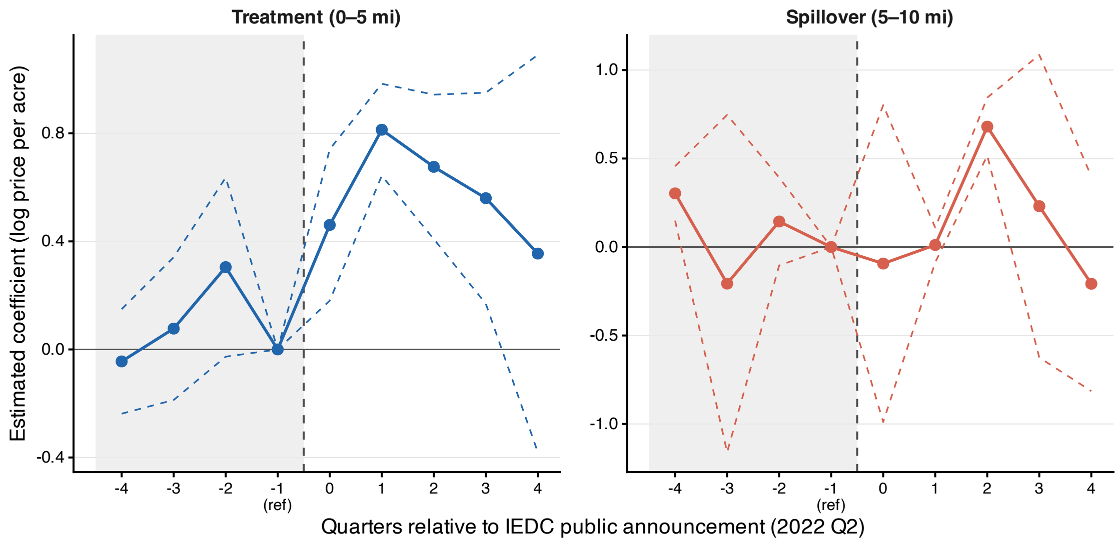
[9]Computed as \$9,331 × ($e^{0.139} - 1$) ≈ \$1,390 per acre, using the same comparison ring baseline.

tations about future infrastructure, related commercial activity, or broader regional development pressure, but they are less directly connected to the land assembly process than parcels within five miles. Additionally, sellers who received above-market proceeds from IEDC may have re-entered the as buyers under Section 1031 like-kind exchange provisions, further transmitting elevated reservation prices into the spillover zone. The pre-announcement indicators for the spillover ring are small and statistically indistinguishable from zero across all specifications, consistent with the parallel trends assumption.

The event-study results provide further insight on both identification and timing of the effects. The parameters are estimated using Weighted Least Squares (WLS) rather than OLS, weighting observations by total acres transacted in each ring-year cell.[10] Figure 4 presents the estimated quarterly coefficients for the treatment (0-5 miles) and spillover (5-10 miles) rings relative to the comparison ring, with quarter -1 as the omitted reference period. The pre-announcement period estimates provide no clear evidence of systematic differential price trends prior to the shock. Most pre-period coefficients are statistically indistinguishable from zero for both rings , supporting the parallel-trends assumption underlying the difference-in-differences design. The treatment ring has a positive point estimate in quarter -2, but this estimate is not statistically significant. We therefore do not interpret it as evidence of a pre-announcement treatment effect. Following the public announcement, the coefficient in 0-5 mile ring coefficient rises sharply and becomes statistically significant, peaking in the first post-announcement quarter at approximately 0.81 log points before moderating to about 0.37 log points by quarter +4. This pattern suggests rapid capitalization of LEAP-related development expectations into nearby farmland values after the project became public. The 5–10 mile ring shows a less immediate and smaller response, with the post-announcement coefficient peaking around quarter +2 before declining. This pattern is consistent with a weaker and more gradual price response outside the core acquisition area.

[10] It is common practice to use weighted estimation when studying dynamic treatment effects in thin asset markets, where transaction intensity varies substantially over time, to address heteroskedasticity and ensure that estimates are not driven by sparsely traded observations (see Case and Shiller, 1987; Amihud, 2002; Solon et al., 2015).

Figure 4: Weighted Event Study: (±4 Quarters)



Notes: The figure plots weighted event-study estimates for the localized treatment region (0–5 miles) and spillover region (5–10 miles) relative to the 10–25 mile comparison group. Coefficients are measured relative to quarter -1, the quarter immediately before the public LEAP announcement in 2022Q2. The estimates are weighted by total acres transacted in each ring-quarter cell.

5.1 Regional Development Shocks versus Urban Gradients

To distinguish the regional development demand shock from ordinary urban-fringe capitalization, we compare the localized estimate with prior development-pressure benchmarks and with an Indiana-specific urban-proximity benchmark estimated over the same sample period. Prior farmland valuation studies show that non-agricultural development expectations are capitalized into farmland prices, but these estimates generally reflect gradual, equilibrium development pressure. The IEDC LEAP District represents a fundamentally different type of event. Rather than gradual urban development pressure, the LEAP District represents a discrete, large-scale regional development demand shock: a single institutional buyer assembling over 9,000 acres at an estimated cost approaching $1 billion, making it one of the largest state-led farmland acquisitions in recent U.S. history (Arnolt Center for Investigative Journalism, 2025).

Prior studies provide some directly relevant equilibrium benchmarks. Using a structural option-value model, Plantinga et al. (2002) estimate that approximately 8 percent of Indiana farmland values reflect expectations of future land development. Livanis et al. (2006) find an urban accessibility elasticity of 0.19 using a generalized spatial three-stage least squares model on U.S. county data, implying that a 10 percent increase in urban accessibility raises farmland values by roughly 1.9 percent. Borchers et al. (2014) find that a 1 percent increase in the share of land facing immediate development potential raises cropland values by 0.43 percent. These estimates are useful benchmarks in that they capture how farmland values are associated with gradual development pressure under normal market conditions. However, these estimates capture how farmland prices respond to gradual development pressure under normal market conditions and do not estimate the price effect of a discrete, localized development demand shock of the kind generated by a voluntary land assembly.

To generate a directly comparable benchmark, we estimate a pooled cross-sectional hedonic model using farmland transactions within 25 miles of 13 Indiana cities with populations exceeding 15,000.[11] The specification mirrors the spatial structure of the main design by defining 0–5 mile and 5–10 mile rings from the city boundary[12] relative to a 10–25 mile baseline and includes the full control set, year fixed effects, and city fixed effects. The pooled 0–5 mile ring coefficient is 0.100 log points (p = 0.006), corresponding to a 10.5 percent premium. This estimate is consistent with the equilibrium benchmarks in the prior literature and confirms that gradual urban proximity generates a modest farmland premium in Indiana. However, the magnitude remains modest relative to the 45.5 percent localized premium estimated around the LEAP District. Table 3 summarizes this comparison.

The comparison clarifies the economic distinction between gradual development pressure and a discrete regional development demand shock. This distinction is increasingly relevant as states compete for advanced manufacturing, semiconductor, battery, and clean-energy investments by

[11] The 13 cities are Fort Wayne, Evansville, South Bend, Carmel, Fishers, Hammond, Gary, Muncie, Terre Haute, Columbus, Kokomo, Bloomington, and Noblesville. Indianapolis is excluded as a categorically distinct urban market. All models include distance to Indianapolis as a control. Full regression results are reported in Appendix Table A9.

[12] City boundary polygons were obtained from the U.S. Census Bureau TIGER/Line shapefiles via the `tigris` R package (Walker, 2016).

Table 3: Development-Related Farmland Price Effects: Prior Literature and Present Study

Study	Method	Geography	Estimate
Panel A: Gradual development pressure and urban-fringe benchmarks			
Plantinga et al. (2002)	Structural option-value model	Contiguous U.S. counties (1997)	8% development share[a]
Livanis et al. (2006)	Generalized spatial 3SLS	U.S. counties (1992, 1997)	$\varepsilon = 0.19$[b]
Borchers et al. (2014)	Hedonic price model	Contiguous U.S. (2010)	$\varepsilon = 0.43$[c]
This paper: urban-fringe benchmark	Cross-sectional hedonic	13 Indiana cities (2018–2024)	10.5%*** ($p = 0.006$)[d]
Panel B: Concentrated regional development demand shock			
This paper: LEAP shock	**Spatial DiD**	**Boone County, Indiana (2018–2024)**	**45.5%*****[e]

[a] Share of Indiana farmland value attributable to future development expectations, from a national cross-section of approximately 3,000 U.S. counties using 1997 Census of Agriculture data.

[b] Elasticity of farmland value with respect to urban accessibility. A 10% increase in urban accessibility raises farmland values by 1.9%.

[c] Elasticity of cropland value with respect to the share of land facing immediate development potential, from the USDA Joint Agricultural Survey.

[d] Pooled 0–5 mile proximity premium relative to the 10–25 mile baseline, estimated across 13 Indiana cities using the same control set as the main specification ($\exp(0.100) - 1 = 0.105$).

[e] Spatial DiD estimate for parcels within 5 miles of the LEAP site relative to the 10–25 mile comparison ring after the public announcement ($\exp(0.375) - 1 = 0.455$).

Notes: *** $p < 0.01$; ** $p < 0.05$. Panel A captures equilibrium development-pressure capitalization. The Panel B estimate uses the spatial difference-in-differences design in Section 3.2.

assembling large rural sites for future industrial use. As these projects become more common across the United States, understanding how concentrated regional development demand shocks are capitalized into farmland prices becomes an important question for land markets, rural development, and place-based industrial policy.

5.2 Robustness Check

We conduct a series of robustness checks to assess the sensitivity of our main findings. Two inference concerns arise from the spatial design. First, unobserved shocks to farmland prices may be spatially correlated across nearby parcels, which county-level clustering does not account for. To address this concern, we re-estimate our preferred specification using Conley spatial standard errors (Conley, 1999) with a 40 kilometer bandwidth, approximately equal to the 25-mile study radius. The results are nearly identical to the baseline: the localized treatment effect remains 0.375 log points and statistically significant, and the spillover estimate remains 0.139 log points and statistically significant (Table A2). Second, the 25-mile study area contains only few counties, so county-clustered standard errors may be unreliable due to the small number of effective clusters. To address the second concern, we construct 5-mile geographic grid cells and implement wild cluster bootstrap inference at the grid-cell level using Webb weights (Webb, 2023), yielding 121 spatial clusters. The localized treatment effect remains statistically significant. The spillover estimate is positive but no longer statistically significant, suggesting the 5-10 mile spillover result is more sensitive to the choice of inference unit and should be interpreted with more caution than the primary localized effect (Table A3). We also collapse the parcel-level data to a ring-year panel as an aggregation check. Although the ring-year model has only 24 observations (8 years $\times$ 3 rings), the estimated effects remain similar in magnitude to the parcel-level results (Table A4). Because of the small number of observations, we interpret this specification descriptively. However, the similarity in magnitudes suggests that the main pattern is not driven by parcel-level weighting or repeated observations within rings.

To examine whether the estimated effects are specific to the LEAP District, we conduct placebo tests by applying the same concentric ring specification to other locations in Indiana. We first apply the model to two county centroids, Clinton County and Benton County as placebo sites.[13] Neither placebo site has a statistically significant treatment or spillover coefficient (Appendix Table A5). We then extend this model to all eligible Indiana County centroids.[14] The actual LEAP treatment estimate lies in the upper tail of the placebo distribution, with a permutation p-value of 0.042.[15]

[13]Clinton County and Benton County were selected for two reasons. First, both are predominantly agricultural and did not experience a comparable industrial or institutional investment shock during 2021–2023. Second, they provide geographic variation: Clinton County is relatively close to the LEAP site while Benton County is distant, spanning different segments of Indiana's farmland market.

[14]Specifically, we exclude Boone County and its contiguous neighbors, and counties with major 2021-2023 industrial or institutional announcements, including Cass County (EDA-funded agribusiness park, 2023) and Lawrence County ($96 million GM electric vehicle casting investment, 2021-2022). County centroids are used as placebo sites because comparable project boundaries do not exist for these locations. This provides a consistent geometric basis for constructing placebo treatment and spillover rings across counties, while acknowledging that the exercise is an approximation rather than a one-to-one replication of the LEAP setting.

[15]The randomization-inference exercise includes 71 placebo sites. Only 3 placebo sites produce a treatment-ring t-statistic as large as the LEAP estimate (t = 5.14).

This result provides additional evidence that the localized price effect is specific to the LEAP regional development demand shock rather than a generic feature of Indiana farmland markets post 2022.

We also conduct checks to examine whether the results are sensitive to alternative definitions of the treatment and comparison groups. First, we drop the 10-15 mile buffer from the comparison group to ensure partially exposed comparison parcels do not bias our estimates downward. After excluding these distance range and using parcels 15-25 miles as the comparison group, the localized treatment effect rises to 0.420 and the spillover effect to 0.166, suggesting that the baseline estimates are conservative (Table A6). Second, instead of assigning treatment using concentric distance rings, we assign treatment at the county level, using Boone County as the treated unit and its four contiguous neighbors as a neighboring-county group. The Boone County coefficient is 0.317 and statistically significant, while the neighboring county coefficient is small and statistically insignificant (Table A7). This result is consistent with most of the 5-10 mile spillover ring falling within Boone county rather than neighboring counties, as shown in Figure 2. Third, we expand the comparison group to all of Indiana. Table A8 presents two columns: a single indicator for parcels within 10 miles of LEAP, and the full ring specification. The single-indicator estimate is 0.231, while separating the localized treatment and spillover rings yields estimates of 0.380 and 0.139, respectively, which are identical to the estimates of our preferred specifications in Table 2. Across all of these specifications, the main localized price effects remain stable, supporting the robustness of the core findings.

6 Conclusion

This paper studies how a concentrated regional development demand shock is capitalized into surrounding farmland markets. We use Indiana's voluntary assembly of more than 8,000 acres for economic development as the empirical setting and find large, persistent, and spatially structured price effects that substantially exceed what equilibrium estimates of gradual development pressure would predict. This evidence suggests that voluntary land assembly, increasingly relied upon by state development agencies across the United States, affects more than the parcels that are directly purchased.

Our main findings are threefold. First, the development demand shock generates a large and highly localized capitalization effect. Parcels within 5 miles of the assembly site appreciated by approximately 45.5 percent, roughly $4,246 per acre relative to the comparison group following the announcement. Farmland in the 5–10 mile spillover ring appreciated by approximately 14.9 percent, or $1,390 per acre, suggesting that price effects may extend beyond the core acquisition zone but decay with distance, although this evidence is more sensitive to alternative inference procedures. Second, event-study estimates provide no clear evidence of differential pre-announcement trends in either ring, followed by a sharp increase in localized price premiums after the public announcement. Third, we further show that this development premium is more than four times larger than the urban-fringe proximity premium estimated from a pooled cross-section of Indiana cities. The contribution of this paper is therefore not simply to document another form of development pressure, but to show how a non-marginal, discrete regional development demand shock can rapidly and substantially reprice nearby farmland.

These findings are consistent with three channels described in the conceptual framework in Section 3.1. The direct demand effect from above-market purchases from development agencies reset local price benchmarks for subsequent arm's-length transactions. The public announcement triggers revaluation of land price as landowners and farmers revise their expectations about future land conversion. Similarly, the presence of a well-capitalized buyer with clear development objectives strengthens sellers' bargaining positions in the market, driving reservation prices upward through the holdout mechanism. The positive 5–10 mile estimate is roughly one-third of the localized treatment premium, suggesting that development-related expectations may extend outward from the shock site, although this spillover evidence is less robust than the core localized effect.

The findings have important implications for land-use and place-based industrial policy. Although voluntary land assembly can attract large-scale investment and generate employment, it can also inflate farmland values and distort market dynamics. When land values rise because of non-agricultural development expectations rather than agricultural fundamentals, farmers face higher acquisition costs even if they do not directly benefit from the development. This appreciation can act as a barrier for local farmers, increase acquisition costs for those seeking to expand operations, and may alter long run expectations about farmland as an asset class (Ahearn and Newton, 2009). The spillover estimates in the 5–10 mile ring further suggest that these effects may extend beyond the immediate shock zone, further complicating land-use planning and rural development decisions. Project-level cost-benefit analyses that focus on acquired parcels, jobs, and infrastructure investment may therefore understate the broader land-market consequences of regional development policy. As states compete for industrial development through voluntary land assembly, policymakers must weigh these benefits against effects on rural land markets and consider how the costs of development are distributed across landowners, farmers, and other rural stakeholders.

Several questions remain for future research. This study focuses on a single development demand shock in one state, and while the LEAP District is large and well-documented, the external validity of the estimated price effects to differently structured shock remains an open question. Future work comparing price effects across multiple voluntary land assemblies like Michigan's Mundy megasite or Tennessee's Memphis Regional Megasite would help identify whether the magnitudes documented here reflect a more general pattern or are specific to this case. Additional research could also examine the mechanisms behind these price effects more directly, as the heterogeneity tests presented here are suggestive but not conclusive. The bargaining channel is especially difficult to assess with administrative transaction records, as final sale prices do not reveal seller reservation prices, buyer willingness to pay, or negotiation dynamics. Tax records could help examine whether Section 1031 reinvestment behavior contributed to price transmission in the spillover zone. Future studies could use laboratory experiments to examine these difficult to observe dynamics. Finally, future work could examine the distributional impacts of these shocks across different stakeholders, including landowners, tenant farmers, beginning farmers, local governments, and non-selling neighboring parcels.

As voluntary land assembly becomes a prominent strategy for industrial and economic development across the United States, understanding its consequences for rural land markets becomes increasingly important. This paper provides a spatial identification framework, quantifies price estimates, and establishes an empirical benchmark against which future voluntary land assemblies can be compared.

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Appendix

Table A1: Suggestive Mechanism Heterogeneity

	(1) Near IEDC Purchase	(2) IEDC-Similar Parcel	(3) Corporate Buyer
Treatment × Post	0.3377***	0.3174***	0.2353**
	(0.0781)	(0.0721)	(0.0858)
Spillover × Post	0.1388**	0.1379**	0.1247**
	(0.0585)	(0.0582)	(0.0506)
Treatment × Post × Near IEDC purchase	0.1781**		
	(0.0712)		
Treatment × Post × IEDC-similar parcel		0.4111***	
		(0.0555)	
Treatment × Post × Corporate buyer			0.3531***
			(0.0690)
County fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	1,196	1,196	1,196
R^2	0.151	0.152	0.138

Notes: All specifications include the same parcel, land-quality, buyer, and locational controls as the preferred difference-in-differences specification. Standard errors are reported in parentheses. The near-IEDC-purchase indicator equals one for parcels located within one mile of an actual IEDC-acquired parcel. IEDC-similar parcels are defined as parcels with above-median acreage and above-median NCCPI soil-quality score. Corporate buyer equals one for transactions in which the buyer is classified as corporate. These heterogeneity estimates should be interpreted as suggestive rather than definitive. None of the proxy variables exclusively captures its proposed mechanism.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2: Robustness to Alternative Inference: Conley Spatial Standard Errors

	(1) Clustered (County)	(2) Conley (40 km)
Treatment × Post	0.375***	0.375***
	(0.073)	(0.070)
Spillover × Post	0.139**	0.139**
	(0.059)	(0.068)
Observations	1,196	1,196
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

Notes: Both columns report estimates from the preferred specification (column 5 of Table 2). Column (1) clusters standard errors at the county level. Column (2) uses Conley spatial standard errors with a 40 kilometers (approximately 25 miles) bandwidth, matching the radius of the study area. Controls include parcel characteristics and infrastructure distances; see Table 2 for the full list. $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Spatial Grid-Cell Wild Bootstrap Inference

	(1) 5-mile grid	(2) 10-mile grid
Treatment × Post	0.375***	0.375***
	(0.112)	(0.112)
Spillover × Post	0.139	0.139
	(0.113)	(0.105)
Grid-cell clusters	121	39
Observations	1,196	1,196
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Controls	Yes	Yes

Notes: Wild bootstrap p-values are computed using 9,999 replications with Webb weights, clustered by spatial grid cells. The 5-mile and 10-mile grid cells are constructed using latitude-longitude bins corresponding approximately to each distance at Indiana's latitude. All specifications include the same controls as the preferred difference-in-differences model, county fixed effects, and year fixed effects.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Robustness to Ring-Year Aggregation

	(1) Ring × Year Panel
Treatment × Post	0.341**
	(0.065)
Spillover × Post	0.175***
	(0.005)
Observations	24
Year FE	Yes
Ring indicators	Yes

Notes: Parcel-level data are collapsed to a ring × year panel (3 rings × 8 years = 24 observations), weighted by total acres transacted. The dependent variable is the acre-weighted mean of log price per acre within each ring-year cell. Year fixed effects and ring indicators are included. Standard errors are clustered at the ring level. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A5: Placebo Tests: Alternative Sites

	(1) Clinton County (North, ~25 mi)	(2) Benton County (Northwest, ~80 mi)
Treatment × Post	0.040	−0.017
	(0.095)	(0.139)
Spillover × Post	0.200	0.186
	(0.157)	(0.169)
Observations	338	302
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

Notes: Each column applies the main DiD specification to a placebo site using the same 0-5, 5-10, and 10-25 mile distance rings centered on the county centroid. Clinton County (north of Boone County) and Benton County (northwest Indiana) were selected for their agricultural character and absence of comparable institutional land acquisition during 2021-2023. Controls include parcel characteristics and infrastructure distances. None of the coefficients are statistically significant at conventional levels. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A6: Robustness Check: Excluding 10–15 Mile Buffer Zone

	Dependent variable: log(Price per Acre)
A. Treatment effects	
Treatment × Post	0.420***
	(0.074)
Spillover × Post	0.166*
	(0.077)
B. Pre-treatment levels	
Treatment	0.008
	(0.144)
Spillover	0.049
	(0.133)
C. Controls	
log(Acres)	−0.172***
	(0.036)
Soil Quality (NCCPI)	0.000
	(0.004)
Has Improvement	−0.157**
	(0.054)
Percent Cropland	0.096
	(0.149)
Corporate Buyer	0.155**
	(0.051)
Dist. to Indianapolis (mi)	0.004
	(0.003)
Dist. to Substation (mi)	−0.002
	(0.065)
Dist. to Solar Farm (mi)	−0.024*
	(0.011)
Dist. to Transmission (mi)	−0.045
	(0.054)
Dist. to Wind Turbine (mi)	0.004
	(0.009)
Dist. to Railroad (mi)	−0.069
	(0.044)
Dist. to Ethanol Plant (mi)	0.004
	(0.007)
Year FE	Yes
County FE	Yes
Observations	982
R^2	0.13

Comparison group comprises farmland transactions 15–25 miles from IEDC purchase sites; parcels 10–15 miles are excluded as a buffer zone. Standard errors clustered by county in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Robustness Check: County-Level Difference-in-Differences

	Dependent variable: log(Price per Acre)
A. Treatment effects	
Boone County × Post	0.317***
	(0.012)
Neighboring Counties × Post	0.025
	(0.048)
B. Controls	
log(Acres)	−0.219***
	(0.010)
Soil Quality (NCCPI)	0.002**
	(0.001)
Has Improvement	−0.053**
	(0.020)
Percent Cropland	0.286***
	(0.039)
Corporate Buyer	0.201***
	(0.015)
Dist. to Indianapolis (mi)	−0.002
	(0.001)
Dist. to Substation (mi)	−0.002
	(0.013)
Dist. to Solar Farm (mi)	0.000
	(0.003)
Dist. to Transmission (mi)	0.012
	(0.013)
Dist. to Wind Turbine (mi)	−0.004
	(0.003)
Dist. to Railroad (mi)	0.023*
	(0.012)
Dist. to Ethanol Plant (mi)	−0.004*
	(0.002)
Year FE	Yes
County FE	Yes
Observations	17,383
R^2	0.21

Treatment group is Boone County; neighboring counties are Clinton, Hamilton, Hendricks, and Montgomery. Comparison group is all remaining Indiana counties. Sample covers arm's-length farmland transactions from 2018–2024. Standard errors clustered by county in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Robustness Check: All-Indiana Comparison Group

	(1) Near LEAP Only	(2) Distance Bands
A. Treatment effects		
Near LEAP (0–10 mi) × Post	0.231**	—
	(0.083)	
Treatment (0–5 mi) × Post	—	0.380***
		(0.054)
Spillover (5–10 mi) × Post	—	0.139*
		(0.073)
B. Pre-treatment levels		
Near LEAP (0–10 mi)	−0.008	—
	(0.087)	
Treatment (0–5 mi)	—	0.000
		(0.064)
Spillover (5–10 mi)	—	0.023
		(0.094)
C. Controls		
log(Acres)	−0.219***	−0.219***
	(0.010)	(0.010)
Soil Quality (NCCPI)	0.002**	0.002**
	(0.001)	(0.001)
Has Improvement	−0.053**	−0.053**
	(0.020)	(0.020)
Percent Cropland	0.286***	0.286***
	(0.039)	(0.039)
Corporate Buyer	0.201***	0.201***
	(0.015)	(0.015)
Dist. to Indianapolis (mi)	−0.002	−0.002
	(0.001)	(0.001)
Dist. to Substation (mi)	−0.002	−0.002
	(0.013)	(0.013)
Dist. to Solar Farm (mi)	0.000	0.000
	(0.003)	(0.003)
Dist. to Transmission (mi)	0.012	0.012
	(0.013)	(0.013)
Dist. to Wind Turbine (mi)	−0.004	−0.005
	(0.003)	(0.003)
Dist. to Railroad (mi)	0.023**	0.024**
	(0.012)	(0.012)
Dist. to Ethanol Plant (mi)	−0.004	−0.004
	(0.002)	(0.002)
Year FE	Yes	Yes
County FE	Yes	Yes
Observations	17,383	17,383
R^2	0.22	0.22

Comparison group comprises all Indiana farmland transactions beyond the treatment zone. Column (1) uses a single indicator for parcels within 10 miles of the LEAP site. Column (2) uses separate treatment (0–5 miles) and spillover (5–10 miles) rings. Standard errors clustered by county in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Cross-Sectional Urban Proximity Benchmarks: Indiana Cities

	Dependent variable: log(Price per Acre)
0–5 mi ring	0.100***
	(0.035)
5–10 mi ring	0.032
	(0.023)
log(Acres)	−0.228***
	(0.013)
Soil Quality (NCCPI)	0.001
	(0.001)
Has improvement	−0.010
	(0.032)
Cropland share	0.329***
	(0.049)
Corporate buyer	0.183***
	(0.024)
Dist. to Indianapolis (mi)	−0.002*
	(0.001)
Dist. to substation (mi)	−0.020
	(0.020)
Dist. to solar farm (mi)	−0.006
	(0.006)
Dist. to transmission line (mi)	0.009
	(0.022)
Dist. to wind turbine (mi)	−0.008
	(0.004)
Dist. to railroad (mi)	0.055*
	(0.022)
Dist. to ethanol plant (mi)	0.002
	(0.003)
Year FE	Yes
City FE	Yes
Observations	13,245
Adjusted R^2	0.19

Dependent variable is log(price per acre). Baseline category is the 10–25 mile ring from city boundary. Sample restricted to transactions within 25 miles of 13 Indiana cities with population >15,000. City boundary polygons from U.S. Census Bureau TIGER/Line shapefiles (Walker, 2016). Standard errors clustered by county in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.